

TRYPANOSOMA SPECIES	
Salivaria	Stercoraria
T. rhodesiense T. gambiense	T. cruzi

1

TRYPANOSOMA SPECIES	
Salivaria	Stercoraria
T. rhodesiense T. gambiense	T. cruzi


HAT


Chagas disease

2

TRYPANOSOME STAGES

Human African Trypanosomiasis (HAT)

Trypomastigote
Epimastigote

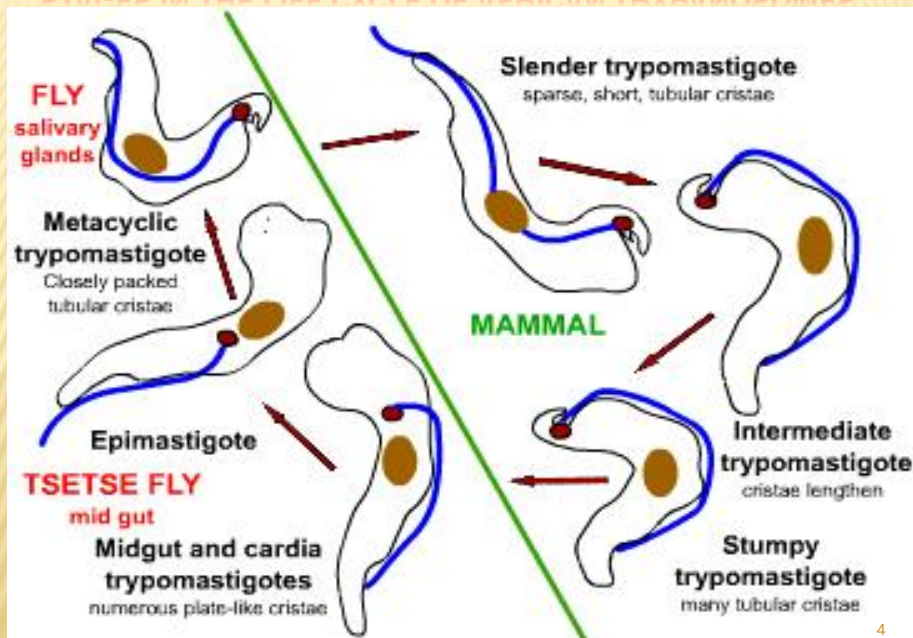
American trypanosomiasis (Chagas Disease)

Trypomastigote
Epimastigote
Promastigote
Amastigote

3

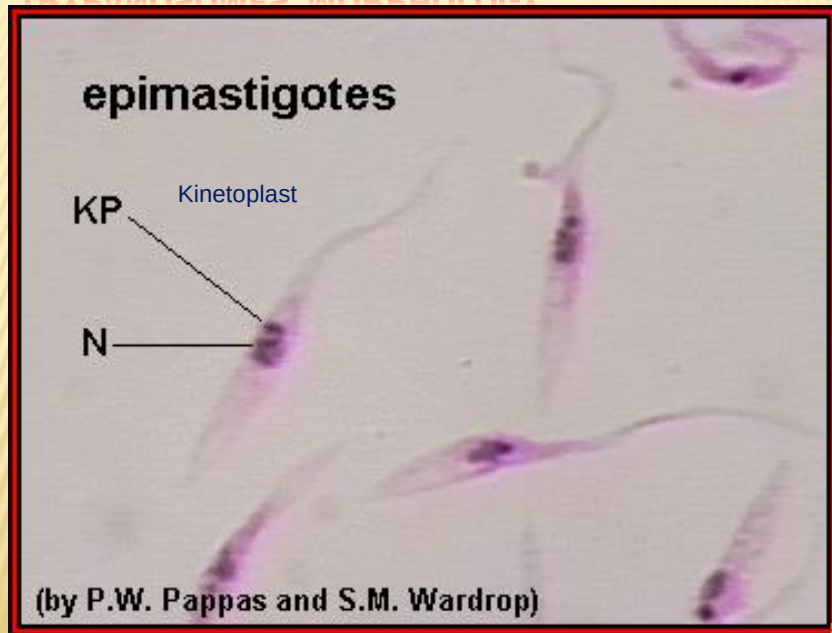
3

STAGES IN THE LIFE CYCLE OF AFRICAN TRYPANOSOMES

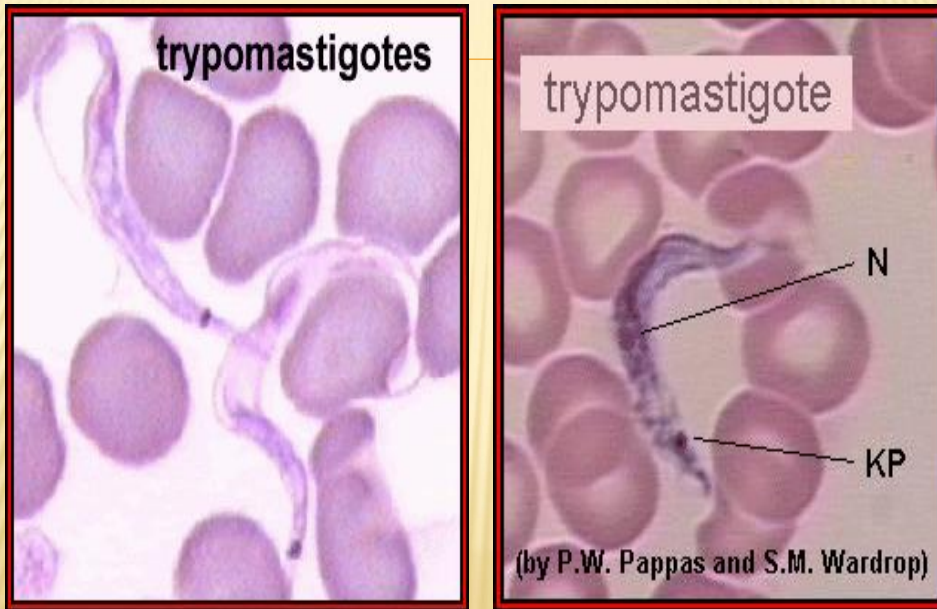


4

4

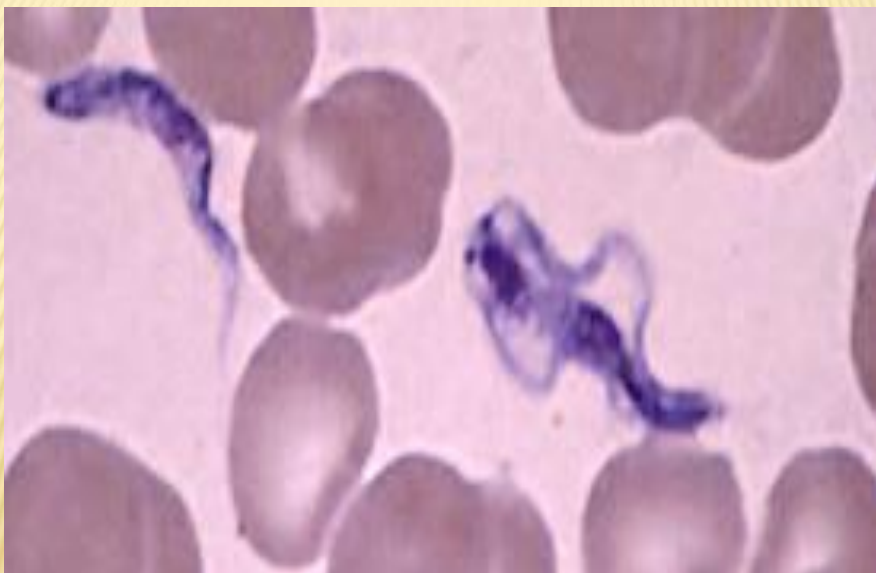
TRYPANOSOMES, MORPHOLOGY

5

TRYPANOSOMES, MORPHOLOGICAL STAGES

6

TRYPANOSOMES, GIEMSA STAINED PBF



7

7

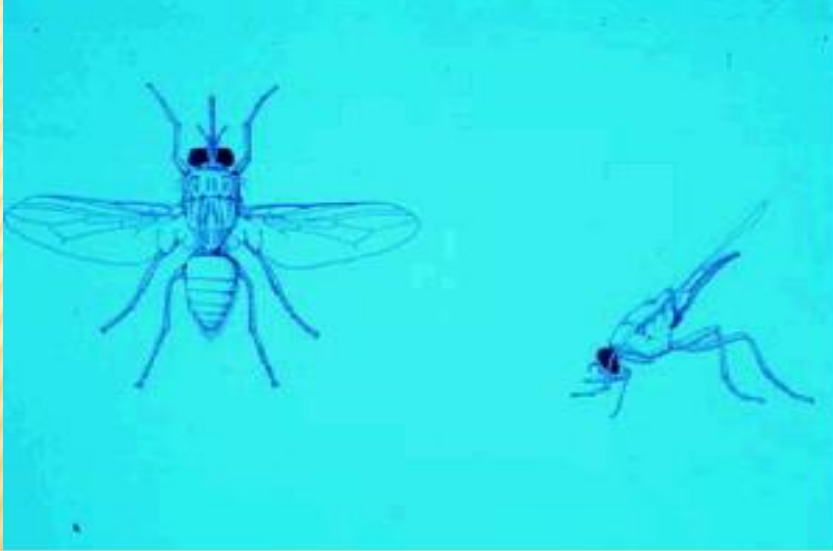
DISTRIBUTION OF AFRICAN TRYPANOSOMES



8

8

VECTORS OF AFRICAN TRYPANOSOMES



Glossina morsitans, a tsetse fly
WHO/TDR/Davies

9

9

VECTORS OF AFRICAN TRYPANOSOMES

Glossina



(by P.W. Pappas and S.M. Wardrop)

10

VECTORS OF AFRICAN TRYPANOSOMES



11

HAT: ANIMAL RESERVOIRS

T. rhodesiense	T. gambiense
Bushbuck Giraffe Hartebeest Hippopotamus Reedbuck Waterbuck Warthog Hyena Lion	Pigs Dogs Cattle Sheep Kob Hartebeest Anthroponosis

12

12



13



14



15



16

NEUROLOGICAL COMPLICATIONS



17

MYXOEDEMA



18

SLEEPING SICKNESS, LATE STAGE: WASTING & SKIN DAMAGE



19

19

HAT: stage determination (disease stages)

Hemo-lymphatic stage (1st stage)

- = Dissemination & proliferation of trypanosomes
- = Lymph and blood

Meningo-encephalitic stage (2nd stage, neurological)

- = After weeks (*T.b.r.*) or months/years (*T.b.g.*)
- = Central nervous system invasion
- = Evolution to coma and death

20

20

Stage –related signs/symptoms

Hemo-lymphatic stage

Fever
Hepatosplenomegaly
Lymphadenopathy

Meningo-encephalitic stage

Daytime somnolence
Nocturnal insomnia
Apathy
Abnormal movements
Psychiatric problems

21

21

Diagnosis and staging

- 1) Blood: Thick films
 Buffy coat
 Capillary centrifugation
- 2) CSF - Lumbar puncture (parasites, WBC counts)



- 3) Lymph node: aspirations

22

22

Treatments for HAT

- Stage-dependent, species-dependent
- Drugs are expensive, toxic
- Currently available drugs:
 - Melarsoprol
 - Eflornithine
 - Nifurtimox
 - Pentamidine
 - Suramin
- Often drug combinations are used
e.g., NECT (Nifurtimox-Eflornithine)
- New drug (oral): Fexinidazole [for T.g]

23

23

Treatments in HAT

	T. rhodesiense	T. gambiense
Hemo-lymphatic (Stage 1)		Pentamidine
	Suramin	Suramin
		Fexinidazole
Meningo-encephalitic (Stage 2)		Eflornithine
	Melarsoprol	Melarsoprol
		Fexinidazole

24

24